Tejas Venu

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Education

University of Virginia — *Aug 2025 – Dec 2026*
Master of Science in Computer Science — Charlottesville, VA
GPA: 4.00 / 4.00

Visvesvarya Technological University — *Dec 2021 – Jul 2025*
Bachelor of Engineering in Computer Science and Engineering — Bengaluru, Karnataka
GPA: 3.90 / 4.00

Work Experience

Internship - AI/ML Engineer — *Feb 2025 – Jul 2025*
Oracle — Bengaluru, Karnataka

- Automated extraction of key legal clauses from **100+ page** documents using **LLMSherpa**, **Word2Vec** embeddings, and **dense vector search** across **8 US regions**.
- Improved retrieval accuracy from **61% to 93%** by designing an **LLM-based taxonomy filtering layer** (Cohere Command R+), eliminating irrelevant semantic matches at scale.
- Led a **5-member** team to deploy the system across diverse document structures, reducing annual client costs from **$2.3M** to **$200K–$300K**.

Internship - Data Analyst — *Jun 2024 – Sep 2024*
Toyota Kirloskar Auto Parts — Bengaluru, Karnataka

- Automated Excel-based safety validations for large datasets (**1000 rows × 700 columns**) using **Python** and **Pandas**, replacing manual checks driven by frequently changing criteria sheets.
- Reduced execution time from **140s to 20s** by eliminating Excel Macros and implementing an optimized Python pipeline.
- Built a **Tkinter** GUI for one-click execution, enabling fully automated validation for gears, axles and driveshafts with **100% accuracy** and zero manual intervention.

Projects

Explainable Skin Lesion Classification with Attention-Enhanced CNNs | Python, PyTorch, EfficientNet-B0 |

- Built an interpretable skin-lesion classification model using **EfficientNet-B0** with a **metadata-conditioned spatial attention** module, embedding explainability directly into training.
- Achieved **87.4%** test accuracy and improved macro-F1 from **0.776 → 0.789**, delivering balanced performance across **7 lesion classes** on the **HAM10000** dataset.
- Replaced post-hoc **Grad-CAM** with **trainable attention** and validated robustness through ablation studies, demonstrating improved interpretability without sacrificing performance.

Human Intruder Detection System | Python, PyTorch, Raspberry Pi, VGGFace |

- Built a real-time intruder detection system using **YOLO, MTCNN and ResNet**, classifying students, faculty, and intruders from live CCTV feeds.
- Achieved **95%+ accuracy** by combining face recognition with uniform and ID-based classification on an augmented dataset.
- Enabled **low-latency, real-time alerts** through integration with existing CCTV infrastructure, ensuring reliable performance across varied lighting and camera angles.

Skills

Programming: Python, Java, C, C++
AI / ML & Data Science: PyTorch, TensorFlow, Keras, Pandas, NumPy, LangChain, Hugging Face
Web Development: HTML, CSS, JavaScript, NodeJS, React, Streamlit, Tkinter, Django
Databases: PostgreSQL, MySQL
Version Control & Analytics Tools: Git, GitHub, Power BI, Matplotlib, Seaborn